

COMPSCI5079 (M)

Cryptography & Secure Development

Dr. Nguyen Truong

Lecturer (Assistant Professor), SoCS, University of Glasgow

Email: Nguyen.Truong@Glasgow.ac.uk

Office: SAWB 221C, Sir Alwyn Williams Building, 18 Lilybank Gardens



Lecture 3:

Symmetric-key Cryptography Block Ciphers





Lecture 3:

Overview



Outline

1. Feistel ciphers
 - A family of single-key, block substitution ciphers
2. Data Encryption Standard (DES)
 - 64-bit Block cipher, 56-bit key length
 - Insecure for modern applications
 - Still widely used
3. Advanced Encryption Standard (AES)
 - 128-bit Block cipher, different key lengths (128, 192, 256)
 - Replacement for DES



Why is AES necessary?

Chronology of DES Cracking	
Broken for the first time	1997
Broken in 56 hours	1998
Broken in 22 hours and 15 minutes	1999
Capable of broken in 5 minutes	2021

Source: Wikipedia



Advanced Encryption Standard (AES)

- The NIST (US National Institute for Standards and Technology) started a search for a replacement for DES once its deficiencies became apparent.
- The design of AES was performed in public, unlike DES.
 - Entries were invited from around the world.
 - The algorithms and analysis were made public.
 - Comments were invited from any organisation.



The Competition

- MARS from IBM
- RC6 from RSA Security
- Twofish from Counterpane (Bruce Schneier)
- Serpens from Ross Anderson, Eli Bihan, Lars Knudsen (Cambridge University Academic lead team)
- Rjindael from Joan Daemen and Vincent Rijmen (Belgian academics)
- The winner, announced in 2000, was Rjindael.



The Rijndael Algorithm

- AES is based on a design principle known as a Substitution-Permutation network (SP network). AES does not use Feistel ciphers
- Rijndael is an iterated block cipher, meaning that it encrypts and decrypts a block of data by the iteration or round of a specific transformation.
- Similar to DES but used calculations involving polynomials rather than table lookups in S-Boxes.
- Most arithmetic is performed with 8-bit values and uses polynomials mod the irreducible polynomial
$$x^8 + x^4 + x^3 + x + 1.$$
- The multiplication lookup table can be implemented in hardware.



Rjindael Blocks and States

- The algorithm can be used with data lengths of 128, 192 or 256 bits, and also key lengths of 128, 192 or 256 bits.
- We will just consider the (128, 128) versions.
- Other versions are similar but have more rounds.
- In what follows
 - The key K is 128 bits long.
 - Each block S is 128 bits long, i.e. 16 bytes called s_i .
- The algorithms for encryption and decryption are related but different.
- The (128, 128) version of the algorithm has 10 rounds, each with its own 128-bit sub-key $K[i]$.



AES Applications



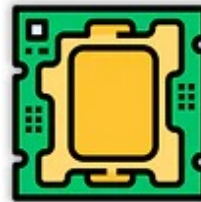
Wireless security



Encrypted browsing



General file encryption

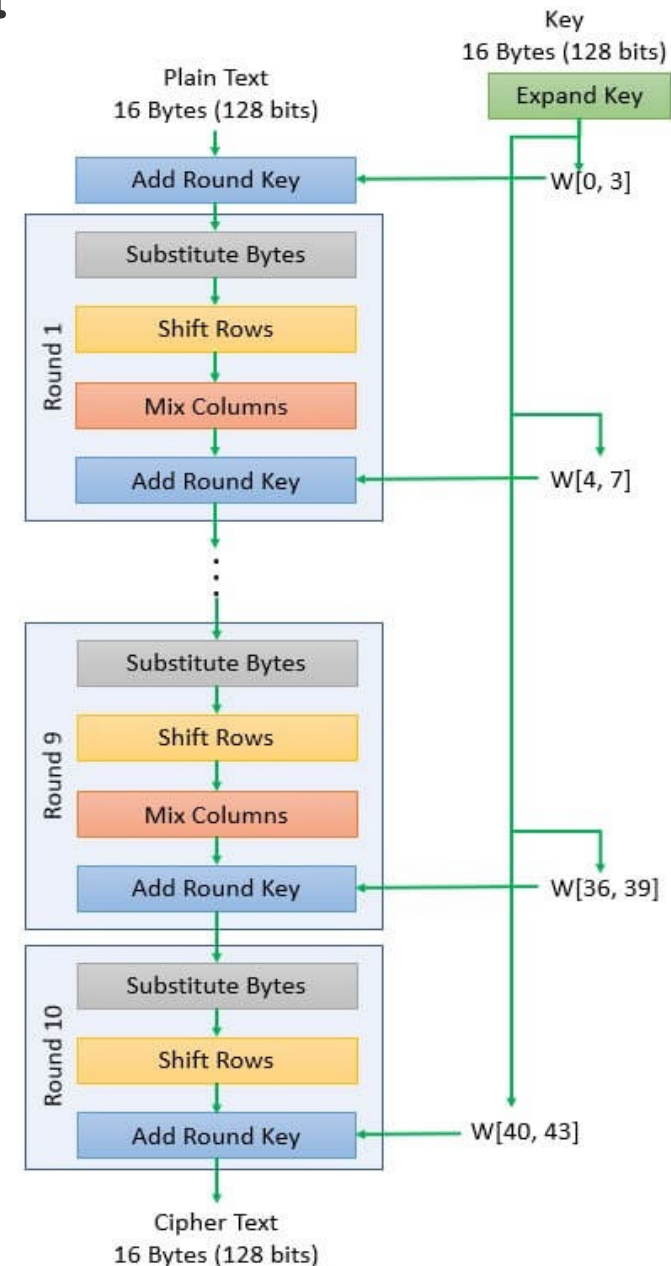


Processor security



AES Encryption Algorithm

- AES operates on a 128-bit plain text block as a single 4X4 matrix which would have a total size of 16 bytes. Every 4 bytes would represent a word.
- The 128-bit key is expanded to form an array containing four 32-bit words. At each round, four distinct words (e.g., W_0 , W_1 , W_2 , W_3) are served to the round key process from the expanded key.
- A round has four functions among which one is of permutation and three are of substitution:
 - Substitute Byte, Mix Column and Add Round Key functions are **substitution** functions
 - Shift Rows is a **permutation** function.
- Only the Add Round key function makes use of the key.





AES Encryption Algorithm

```
AddRoundKey (S, K[0]) ;  
for (int round = 1; round <= 10; round++) {  
    SubBytes (S) ;  
    ShiftRows (S) ;  
    if (round != 10)    //Not final round  
        MixColumns (S) ;  
  
    AddRoundKey (S, K[round]) ;  
}
```

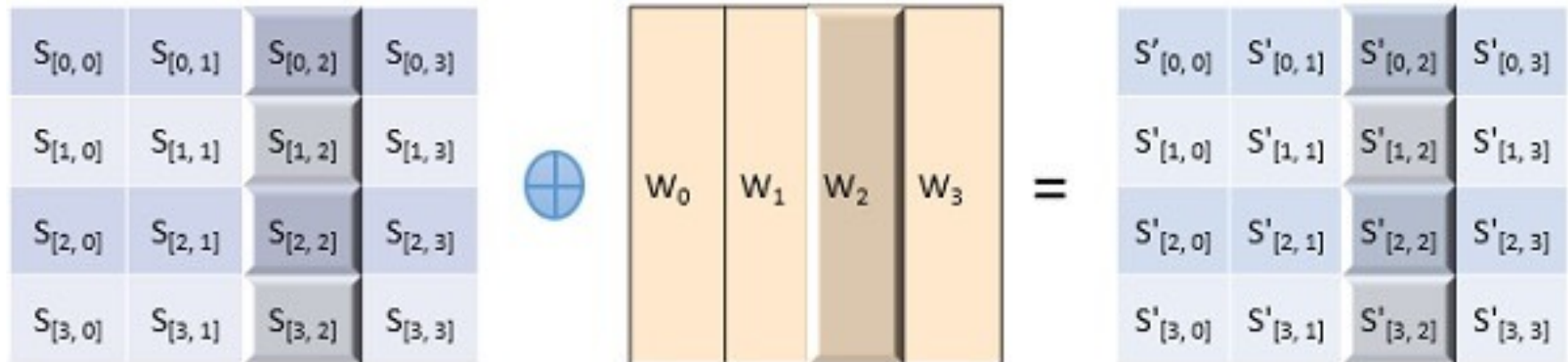


Function: $\text{AddRoundKey}(S, K)$

- This is very simple.
- S and K are XOR'ed bit-by-bit, and the result is left in S .



Function: $\text{AddRoundKey}(S, K)$



Add Round Key



Function: SubBytes (S)

SubBytes function is a simple transform which converts 8-bit data to other 8-bit data:

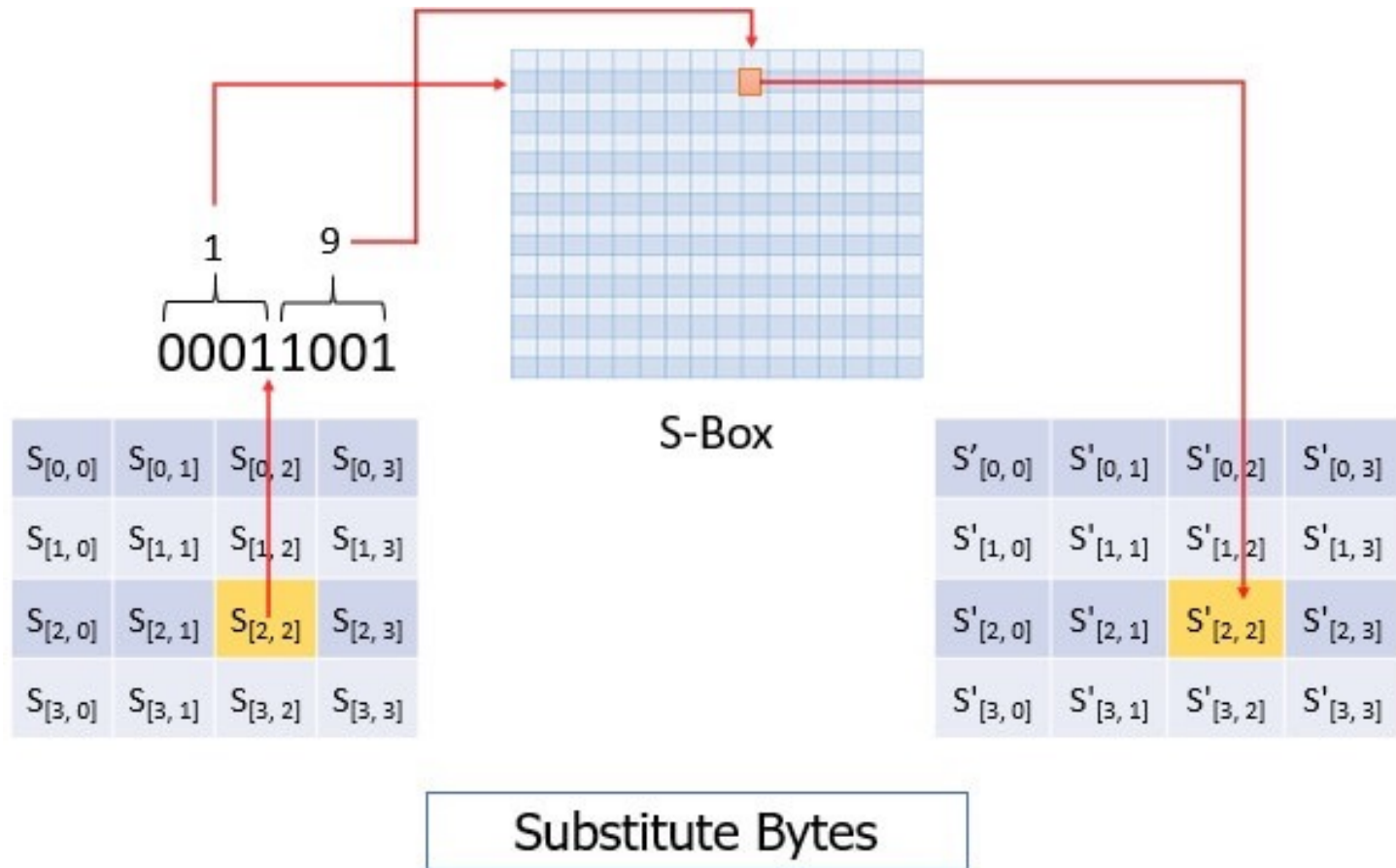
- 8-bit polynomial arithmetic is used.
- s , x , y are all 8-bit values.
- $x = s^{-1}$ where the inverse of 0 is 0.
- $y = Mx$, where the **matrix M** has single-bit entries:

1	0	0	0	1	1	1	1
1	1	0	0	0	1	1	1
1	1	1	0	0	0	1	1
1	1	1	1	0	0	0	1
1	1	1	1	1	0	0	0
0	1	1	1	1	1	0	0
0	0	1	1	1	1	1	0
0	0	0	1	1	1	1	1

- Bitwise arithmetic is used: $+$ / $-$ / XOR.
- y now replaces the old value of s .



Function: SubBytes (S)





The Matrix M

- This simple structure of the matrix M allows a formal proof that the algorithm is resilient to attacks by differential and linear cryptanalysis.
- This contrasts with the choice of S -boxes in the DES algorithm, which looks like they were chosen randomly.



Function: ShiftRows (S)

- The ShiftRows and MixColumns subroutines ensure that all bytes interact with each other in the algorithm.
- They both treat the 16 bytes of the state S as a 4x4 matrix.
- ShiftRows is just a simple shift of the 4 bytes in each row.
- The amount of shift increases with each row.

$S_0 \ S_1 \ S_2 \ S_3 \rightarrow S_0 \ S_1 \ S_2 \ S_3$

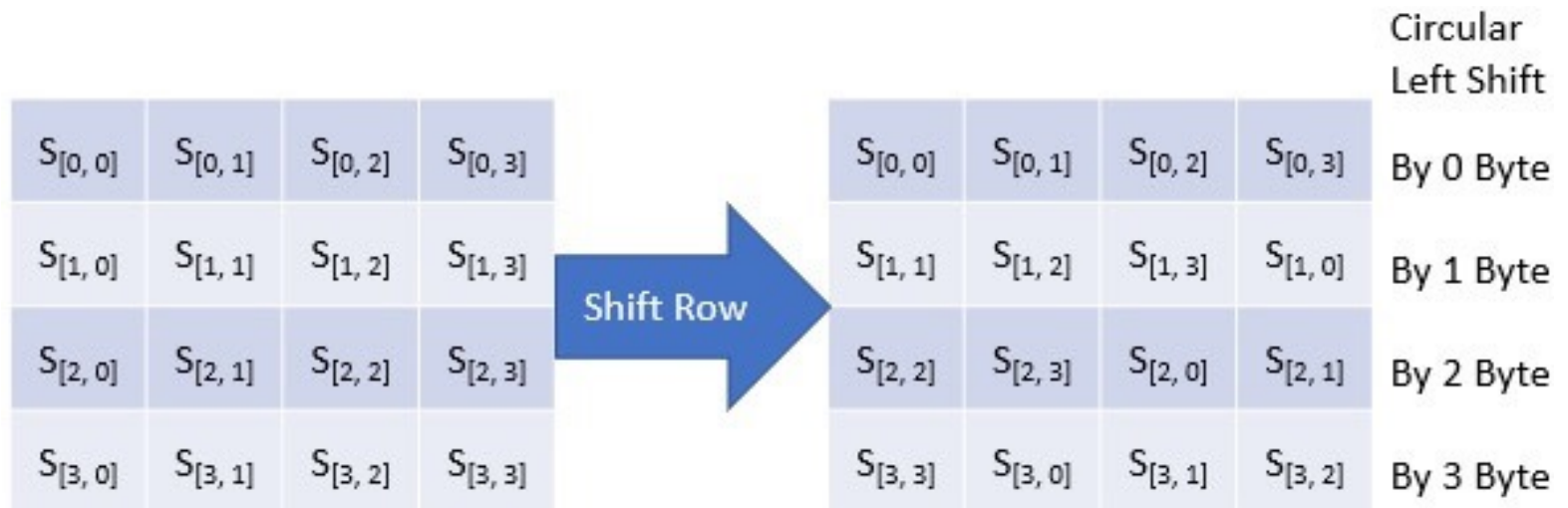
$S_4 \ S_5 \ S_6 \ S_7 \rightarrow S_5 \ S_6 \ S_7 \ S_4$

$S_8 \ S_9 \ S_{10} \ S_{11} \rightarrow S_{10} \ S_{11} \ S_8 \ S_9$

$S_{12} \ S_{13} \ S_{14} \ S_{12} \rightarrow S_{15} \ S_{12} \ S_{13} \ S_{14}$



Function: ShiftRows (S)



Shift Rows



Function: MixColumns (S)

- The 16 bytes can be represented as a 4x4 matrix.
- Each column of 4 bytes is treated as a cubic polynomial in X , with 8-bit coefficients, the polynomial for the 0th column is:

$$s_0 + s_4 X + s_8 X^2 + s_{12} X^3$$

- Each column polynomial is multiplied by a fixed polynomial with 8-bit coefficients which are:

$$\text{col } 0) \quad P_2 + P_3 X + P_1 X^2 + P_1 X^3$$

$$\text{col } 1) \quad P_1 + P_2 X + P_3 X^2 + P_1 X^3$$

$$\text{col } 2) \quad P_1 + P_1 X + P_2 X^2 + P_3 X^3$$

$$\text{col } 3) \quad P_3 + P_1 X + P_1 X^2 + P_2 X^3$$

- Where coefficients are 8-bit polynomials $P_1 = 1$, $P_2 = x$, $P_3 = x + 1$.
 - Naturally, each 8-bit coefficient also uses polynomial arithmetic!
- These polynomials are reduced mod another polynomial $X^4 + 1$
 - This is not irreducible, and so care has to be taken to choose numbers that have inverses.



Function: MixColumns (S)

$$\begin{bmatrix} 02 & 03 & 01 & 01 \\ 01 & 02 & 03 & 01 \\ 01 & 01 & 02 & 03 \\ 03 & 01 & 01 & 02 \end{bmatrix} \times \begin{bmatrix} S_{[0,0]} & S_{[0,1]} & S_{[0,2]} & S_{[0,3]} \\ S_{[1,0]} & S_{[1,1]} & S_{[1,2]} & S_{[1,3]} \\ S_{[2,0]} & S_{[2,1]} & S_{[2,2]} & S_{[2,3]} \\ S_{[3,0]} & S_{[3,1]} & S_{[3,2]} & S_{[3,3]} \end{bmatrix} = \begin{bmatrix} S'_{[0,0]} & S'_{[0,1]} & S'_{[0,2]} & S'_{[0,3]} \\ S'_{[1,0]} & S'_{[1,1]} & S'_{[1,2]} & S'_{[1,3]} \\ S'_{[2,0]} & S'_{[2,1]} & S'_{[2,2]} & S'_{[2,3]} \\ S'_{[3,0]} & S'_{[3,1]} & S'_{[3,2]} & S'_{[3,3]} \end{bmatrix}$$

Mix Columns



AES Decryption

- Inverse operations are performed in the reverse order.
- MixColumns, ShiftRows and SubBytes each have a simple inverse operation that can undo its effects.
- AddRoundKey is its own inverse.
- InverseSubBytes multiplies by the reverse 8-bit matrix, and then calculates the polynomial inverse.
- InverseShiftRows shifts left rather than right.
- InverseMixColumns inverts the column matrix and performs the same calculations.



Sub-key Generation

- We need to produce 10+1 round keys, each of 128 bits, from the initial 128-bit key.
- Similar operations are used to generate the sub-keys.
 - Shifts
 - 8-bit polynomial arithmetic.
 - Exclusive OR.



DES and AES in Comparison

DES Algorithm	AES Algorithm
Key Length - 56 bits	Key Length - 128, 192, 256 bits
Block Size - 64 bits	Block size - 128 bits
Fixed no. of rounds	No. of rounds dependent on key length (10, 12, and 14)
Slower and less secure	Faster and more secure



Summary

- AES was adopted after an open competition.
- It uses many rounds, each with its own sub-key.
- All the operations are fast and don't require much silicon.
- The basic transformations use mathematics so it can be proved to be secure from all known attacks.
- Both the key and block size can be 128, 192 or 256 bits long.



Quizzes

4. Describe how the process of choosing the AES algorithm was different from the one that led to DES. Was the new process better or worse? In what ways is AES better than DES?
5. Describe how the Rijndael AES encryption algorithm uses polynomial arithmetic. What is the advantage of using this arithmetic over the use of S-boxes in DES?